Year 13 Maths - Rational functions and partial fractions

Similar questions - Simplifying and splitting fractions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Simplify fully $\dfrac{x^3-2x^2-5x+6}{x^2-x-6}$. **[4]**

(b) Show that $\dfrac{1}{2-\sqrt{x}}+\dfrac{1}{2+\sqrt{x}}$ can be written in the form $\dfrac{a}{b-x}$, where a and b are constants to be found. **[3]**

(c) Hence solve $\dfrac{1}{2-\sqrt{x}}+\dfrac{1}{2+\sqrt{x}}=5$. **[2]**

2

(a) Express $\dfrac{2x^2+5x-1}{x^2-1}$ in the form $A+\dfrac{B}{x-1}+\dfrac{C}{x+1}$. **[5]**

(b) Hence show that $\displaystyle\int_2^4 \frac{2x^2+5x-1}{x^2-1}\,\mathrm{d}x = 4+\ln 75$. **[4]**

Question 1

(a) Numerator: $x=1$ and $x=3$ and $x=-2$ are roots, so it factorises as $(x-1)(x-3)(x+2)$. Denominator: $(x-3)(x+2)$. The fraction simplifies to $x-1$ (for $x \neq 3, -2$)

(b) $\dfrac{(2+\sqrt{x})+(2-\sqrt{x})}{(2-\sqrt{x})(2+\sqrt{x})} = \dfrac{4}{4-x}$, so $a=4$, $b=4$

(c) $\dfrac{4}{4-x} = 5$, so $4 = 20-5x$ and $x = \dfrac{16}{5}$. This is valid: $x \geqslant 0$ and $x \neq 4$

Question 2

(a) Dividing: $2x^2+5x-1 = 2(x^2-1)+5x+1$, so the fraction is $2+\dfrac{5x+1}{(x-1)(x+1)}$. $\dfrac{5x+1}{(x-1)(x+1)} = \dfrac{B}{x-1}+\dfrac{C}{x+1}$ gives $5x+1 = B(x+1)+C(x-1)$: $x=1$: $B=3$; $x=-1$: $C=2$. So $2+\dfrac{3}{x-1}+\dfrac{2}{x+1}$

(b) $[2x+3\ln(x-1)+2\ln(x+1)]_2^4 = (8+3\ln 3+2\ln 5)-(4+0+2\ln 3) = 4+\ln 3+2\ln 5 = 4+\ln(3\times 25) = 4+\ln 75$